
PIANO MOVERS PROBLEM BY ANT COLONIES

By

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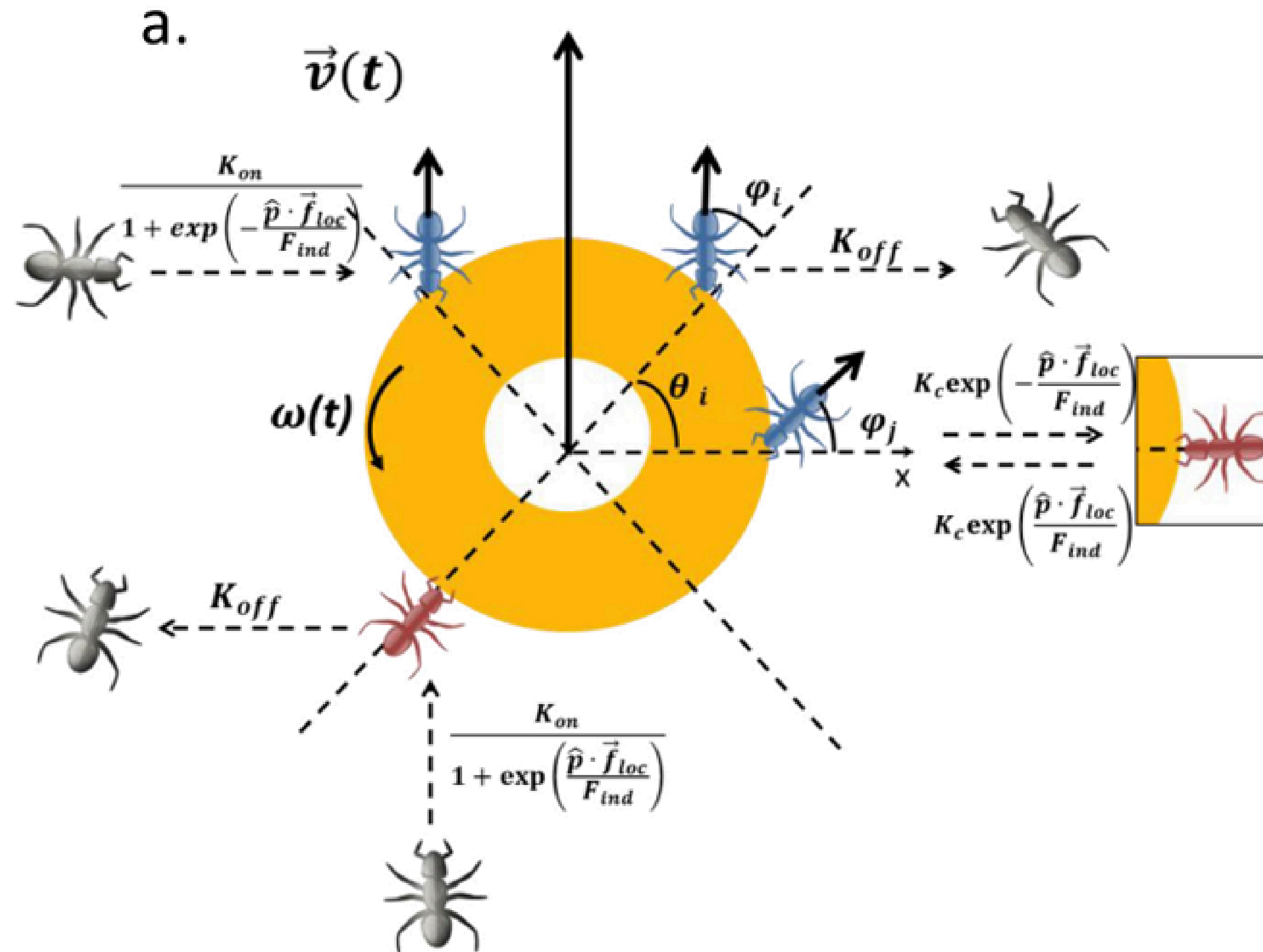
Model Assumptions

- Ants either pull (contribute force to cargo velocity) or lift (reduce friction by factor $\beta \cdot f_0$), with no pushing behavior.
- Cargo motion is overdamped — inertia is neglected; velocity responds linearly to net force at each instant.
- Ants sense only the local force signal at their attachment point (translational + rotational components) to decide roles.
- Each ant has an individuality parameter F_{ind} governing how strongly local cues influence its puller/lifter decision.
- Newly attached ants carry directional information about the nest but forget it stochastically at rate $K_{\text{forget}} = 0.1 \text{ sec}^{-1}$.

System and Ant Description

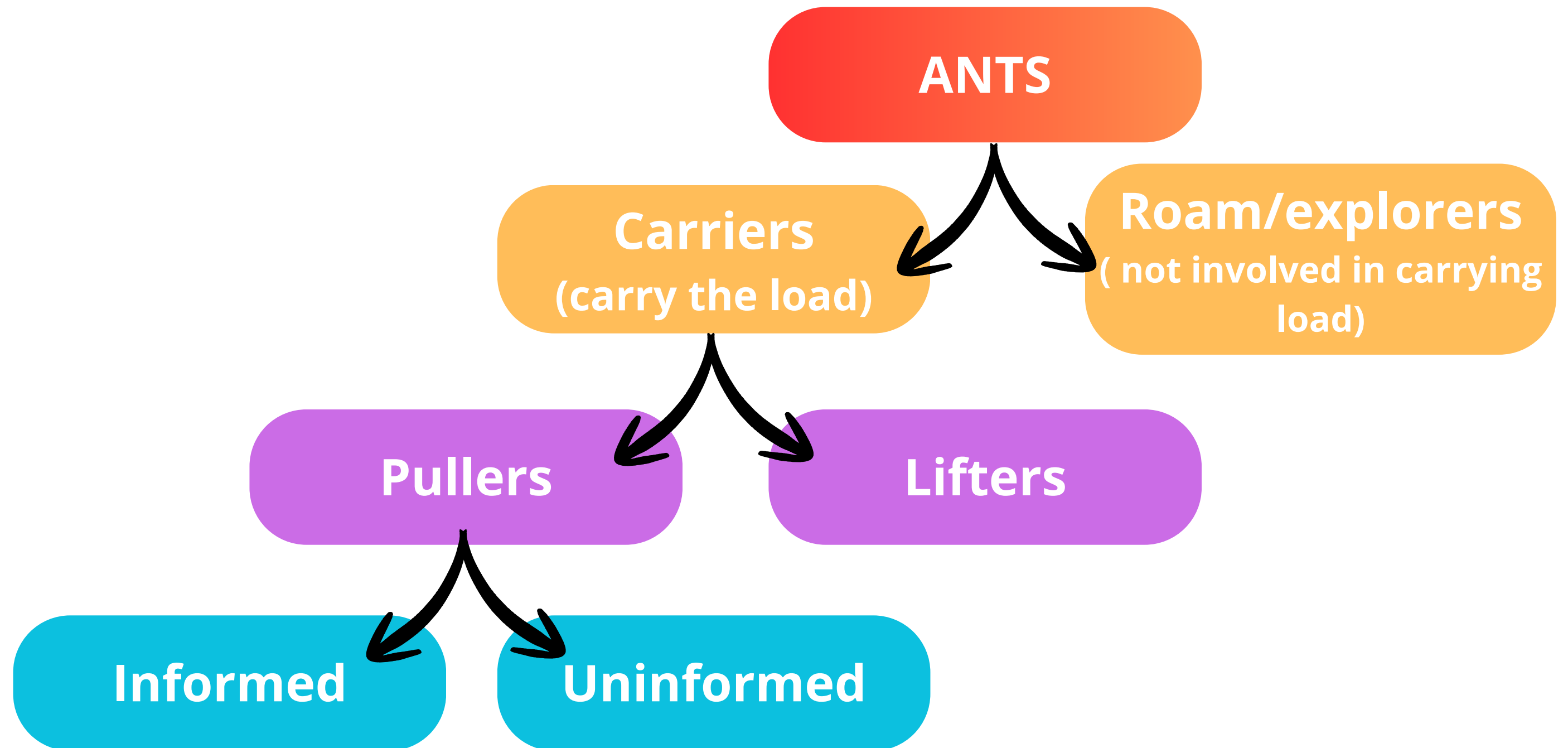
- The cargo is modeled as a ring-shaped object divided into N_{max} equally spaced attachment sites, each occupied by a puller, lifter, or empty.
- A reservoir of N_{av} surrounding ants can attach to empty sites at rate K_{on} and detach at rates K^1_{off} (static cargo) or K^2_{off} (moving cargo).
- Pullers are angularly offset from the radial direction by $\phi_i \in [-\phi_{\text{max}}, +\phi_{\text{max}}]$, allowing the group to generate both translational velocity and torque.
- Lifters attach radially ($\phi_i = 0$) and contribute no net force, only friction reduction — they are more prevalent on the trailing edge.
- The collective direction emerges from a mesoscopic order-disorder transition analogous to an Ising ferromagnet, where group size and F_{in_d} jointly determine whether motion is persistent or random-walk-like.

System and Ant Description



The Model

The ants can either pull the cargo along their body axis, or lift the cargo to reduce its friction with the surface - **Pullers and Lifters**



Pullers - Driving the cargo Forward

What they do: Pullers are attached to the cargo and actively exert force to move it. They are angularly offset from the radial direction by ϕ_i , allowing them to pull the cargo both translationally and rotationally. They dominate the leading edge of the cargo and are the primary drivers of directed motion.

$$V_{\text{cm},x} = \frac{f_0 \sum_{i=1}^{N_{\text{max}}} n_p(\theta_i, \varphi_i) \cos(\theta_i + \varphi_i) - f_{\text{kin},x}}{\gamma}$$
$$V_{\text{cm},y} = \frac{f_0 \sum_{i=1}^{N_{\text{max}}} n_p(\theta_i, \varphi_i) \sin(\theta_i + \varphi_i) - f_{\text{kin},y}}{\gamma}$$
$$\omega = \frac{f_0 \sum_{i=1}^{N_{\text{max}}} n_p(\theta_i, \varphi_i) \sin(\varphi_i) - \tau_{\text{kin}}}{\gamma_{\text{rot}}}$$

where $n_p(\theta_i, \varphi_i) = 1$ only if site i is occupied with a puller

Constant	Meaning
f_0	Force magnitude of a single puller ant – sets the per-ant contribution scale
$n_p(\theta_i, \phi_i)$	Occupancy operator – 1 only if site i has a puller, else 0
$\gamma = M/\delta t$	Cargo response coefficient – how quickly cargo accelerates per unit net force; larger γ means heavier/slower cargo
$\gamma_{\text{rot}} \approx 0.4 \cdot \gamma$	Rotational response – derived from ring geometry ($a=0.36$, $b=0.57$ cm); resists angular acceleration
$f_{\text{kin}}, \tau_{\text{kin}}$	Kinetic friction force and torque opposing translation and rotation respectively
$\phi_i \in [-\phi_{\text{max}}, +\phi_{\text{max}}]$	Angular offset of puller from radial – $\phi_{\text{max}} = 52^\circ$ fitted from experiment; allows torque generation

Lifters — Reducing Ground Friction

What they do: Lifters attach radially to the cargo ($\phi_i = 0$) and do not contribute any net directional force. Instead they partially lift the cargo off the surface, reducing the effective friction the pullers must overcome. They tend to accumulate on the trailing edge since the local force signal there disfavors pulling.

Governing Equation:

Effective friction reduction: $f_{\text{friction}} \rightarrow \beta \cdot f_0 \cdot N_{\text{lifters}}$

Lifter radial alignment means $\sin(\phi_i) = 0 \rightarrow$ zero torque contribution

Puller/lifter probability (P_p and P_l) respectively at attachment:

$$P_p = \frac{1}{1 + \exp\left(-\frac{\mathbf{p}_i \cdot \mathbf{f}_{\text{loc}}}{F_{\text{ind}}}\right)}$$
$$P_l = 1 - P_p = \frac{1}{1 + \exp\left(\frac{\mathbf{p}_i \cdot \mathbf{f}_{\text{loc}}}{F_{\text{ind}}}\right)}$$

Constant	Meaning
$\beta = 1.65$	Friction reduction factor per lifter — fitted from simulation; >1 means each lifter reduces friction by more than one ant-force unit
f_0	Force scale of a single ant — same unit used for both pulling force and friction reduction
P_l	Probability of becoming a lifter — increases when local force signal opposes the site's outward direction
F_{in_d}	Individuality parameter — large F_{in_d} makes lifter/puller choice insensitive to local force, small F_{in_d} makes it deterministic
$p_i \cdot f_{loc}$	Dot product of site polarization with local force — negative value on trailing edge drives lifter decision
$f_{stat} = 2.7$ ant-forces	Static friction threshold — lifters collectively help the group exceed this to initiate motion

Newly Attached (Informed) Ants — Injecting Directional Information

What they do: Ants that just attached were recently free-moving and carry knowledge of the nest direction. In the first ~10 seconds after attachment they steer the cargo by pulling persistently toward the nest from the leading edge. Their influence decays stochastically as they lose this knowledge, behaving like uninformed ants thereafter.

Governing Equations :

- Informed pulling toward nest: ant orients toward +x (nest direction) regardless of f_{loc}
- Forgetting rate: $K_{forget} = 0.1 \text{ sec}^{-1} \rightarrow$ mean informed duration = 10 sec
- Role switching rates while informed:

$$R_{l \rightarrow p} = K_c \exp \left(\frac{\mathbf{p}_i \cdot \mathbf{f}_{loc}}{F_{ind}} \right)$$
$$R_{p \rightarrow l} = K_c \exp \left(-\frac{\mathbf{p}_i \cdot \mathbf{f}_{loc}}{F_{ind}} \right)$$

Constant	Meaning
$K_{\text{forget}} = 0.1 \text{ sec}^{-1}$	Rate of losing nest-direction knowledge — gives ~10 sec window of directed influence
$K_c = 0.7 \text{ sec}^{-1}$	Basal rate of role conversion and reorientation — controls how fast informed ant recruits others
$t_{\text{forg}} = 10 \text{ sec}$	Forgetting timescale in gradual model — exponential decay of informed vs. uninformed state probability
f^i_{loc}	Local force signal — informed ant overrides this and orients to +x, bypassing normal signal
$F_{\text{ind}} = 10 \text{ ant-forces}$	Sets how strongly the informed ant's pulling biases neighboring ants' role decisions
$N_{\text{av}} = 11$	Mean surrounding ants in clean-board setting — determines how many candidates can be recruited by informed ant's signal

Gillespie Events - Stochastic time Evolution



1 Attachment: $R_{\text{att}} = K_{\text{on}} \cdot N_{\text{av}} \cdot N_{\text{empty}}$

2 Forgetting: $R_{\text{forget}} = k_{\text{forget}} \cdot N_{\text{informed}}$

3 Detachment: $R_{\text{det}} = N_{\text{att}} (K_{\text{off}}^1 \delta(f_{\text{loc}}) + K_{\text{off}}^2 (1 - \delta(f_{\text{loc}})))$

4 Conversion: $R_{\text{con}} = K_c \sum_{i=1}^{N_{\text{max}}} \left(n_i^{\text{p}} \exp \left(-\frac{\mathbf{p}_i \cdot \mathbf{f}_{\text{loc}}^i}{F_{\text{ind}}} \right) + n_i^{\text{l}} \exp \left(\frac{\mathbf{p}_i \cdot \mathbf{f}_{\text{loc}}^i}{F_{\text{ind}}} \right) \right)$

5 Reorientation: $R_{\text{orient}} = K_c N^{\text{p}}$

$$R_{\text{tot}} = R_{\text{att}} + R_{\text{det}} + R_{\text{con}} + R_{\text{orient}}$$

This is the time step taken to update the dynamics of the cargo!!

rnd is drawn from a uniform distribution- U(0,1)

$$x_{\text{cm}}(t + dt) = x_{\text{cm}}(t) + V_x(t)dt$$

$$y_{\text{cm}}(t + dt) = y_{\text{cm}}(t) + V_y(t)dt$$

$$dt = -\frac{1}{R_{\text{tot}}} \log(\text{rnd})$$

Cargo Velocity and angular velocity update

$$\mathbf{V}_{\text{cm}} = \frac{\mathbf{f}_0 - \mathbf{f}_{\text{kin}}}{\gamma}$$
$$\omega = \frac{\tau_0 - \tau_{\text{kin}}}{\gamma_{\text{rot}}}$$

- $\gamma_{\text{rot}} \approx 0.4 \text{ cm} \cdot \gamma$ (derived from ring geometry with inner radius $a=0.36 \text{ cm}$, outer $b=0.57 \text{ cm}$)
- $f_{\text{kin}} \approx 0.9 \cdot f_{\text{stat}}$; $f_{\text{stat}} = 2.7 \text{ ant-forces}$; $\tau_{\text{stat/kin}} = 0.83 \cdot f_{\text{stat/kin}}$

If $\omega = 0$ the cargo rotates and the ants rotate with it. Therefore, their forces also rotate which cause the center of mass velocity to rotate as well.
We divide the dt into many small time steps: $\delta t = 0.01 \text{ sec}$.

$$x_{\text{cm}}(t + \delta t) = x_{\text{cm}}(t) + V_x(t)\delta t$$

$$y_{\text{cm}}(t + \delta t) = y_{\text{cm}}(t) + V_y(t)\delta t$$

$$\theta_i(t + \delta t) = \theta_i(t) + \omega\delta t$$

$$\mathbf{f}_i(t + \delta t) = \mathbf{f}_i(t) + \boldsymbol{\omega} \times \mathbf{f}_i(t)\delta t$$

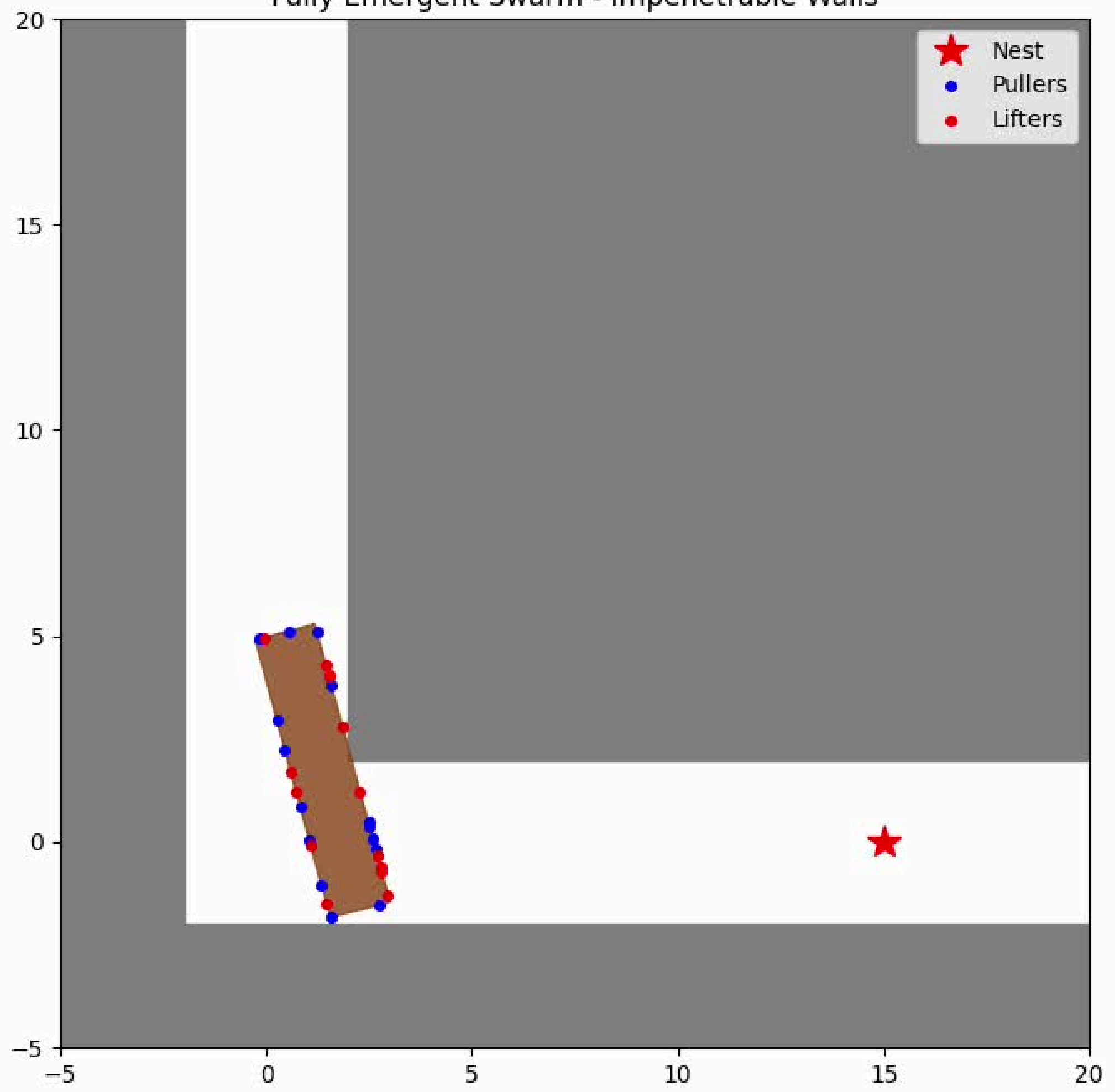
$$\mathbf{v}_{\text{cm}}(t + \delta t) = \mathbf{v}_{\text{cm}}(t) + \boldsymbol{\omega} \times \mathbf{v}_{\text{cm}}(t)\delta t$$

Role Assignment

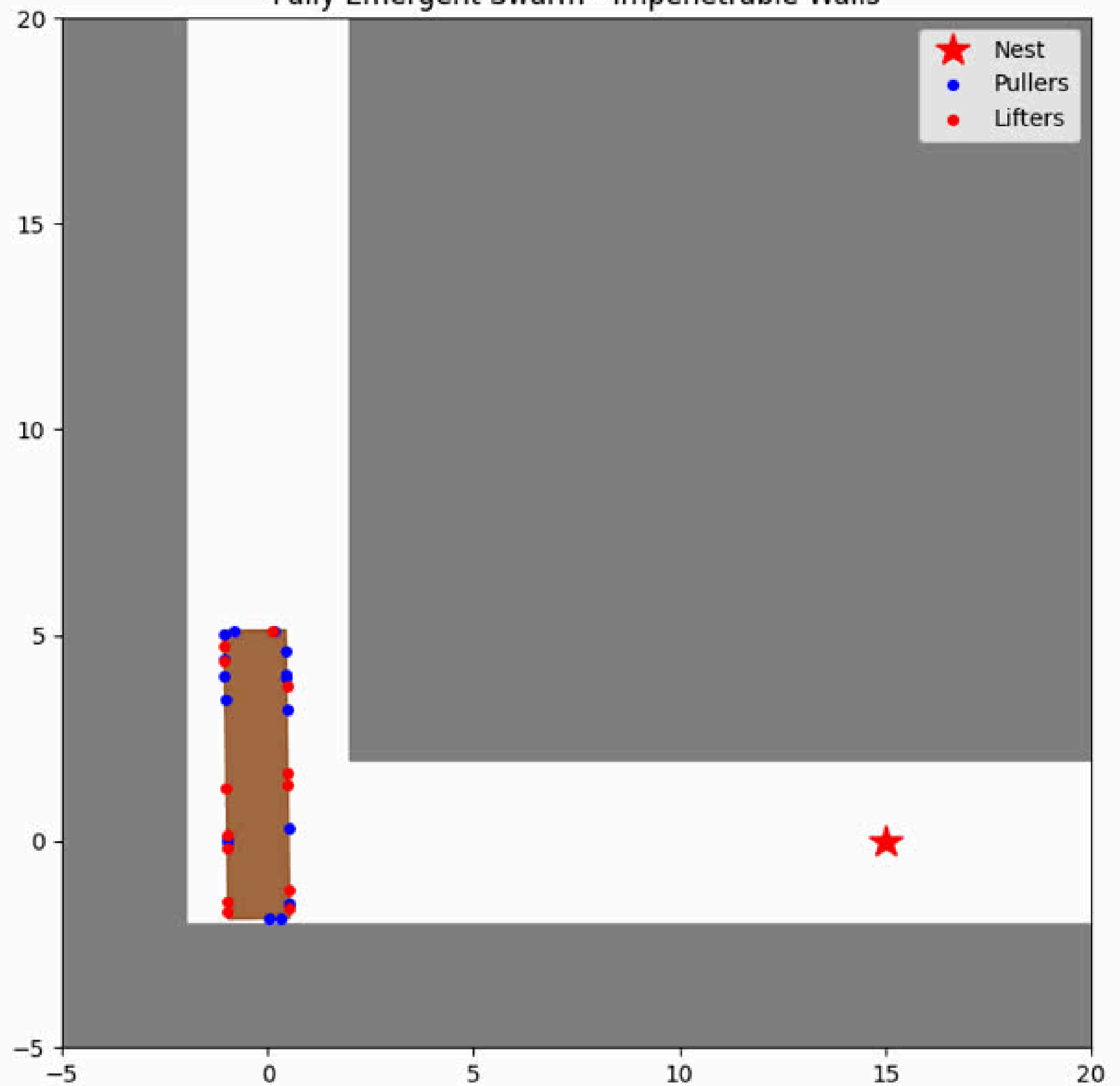
rnd is drawn from a uniform distribution- $U(0,1)$

- $\text{rnd} < R_{\text{att}}/R_{\text{tot}}$: **Attachment** to load
- $R_{\text{att}}/R_{\text{tot}} \leq \text{rnd} < (R_{\text{att}} + R_{\text{det}})/R_{\text{tot}}$: **Detachment** from load
- If $(R_{\text{att}} + R_{\text{det}})/R_{\text{tot}} \leq \text{rnd} < (R_{\text{att}} + R_{\text{det}} + R_{\text{con}})/R_{\text{tot}}$: a **conversion** event (puller to lifter or vice versa) occurs.
- If $\text{rnd} \geq (R_{\text{att}} + R_{\text{det}} + R_{\text{con}})/R_{\text{tot}}$: **Reorientation** - we randomly choose a puller so she aligns with the local velocity.

Rally Emergent Swarm - Impenetrable Walls

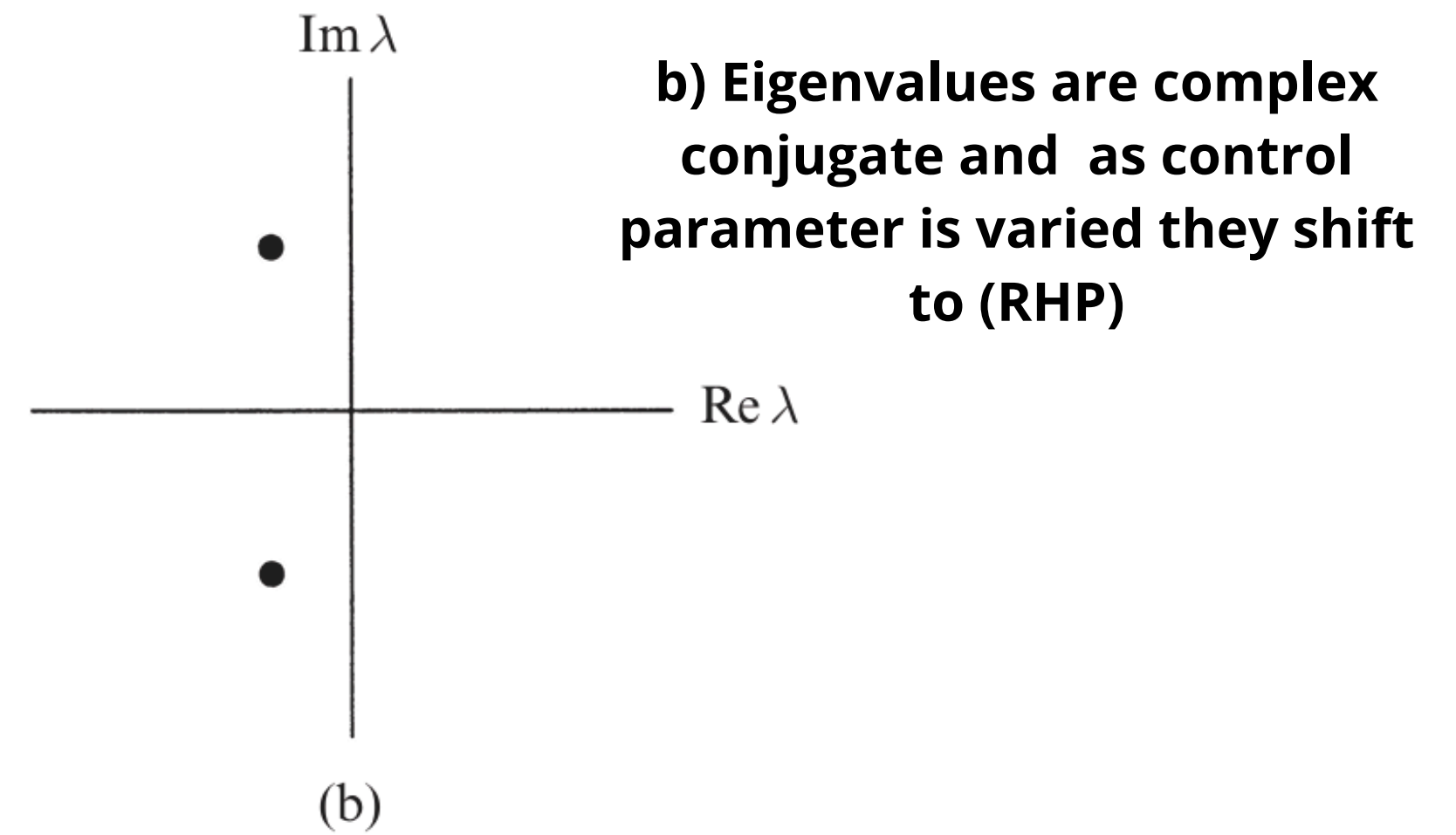
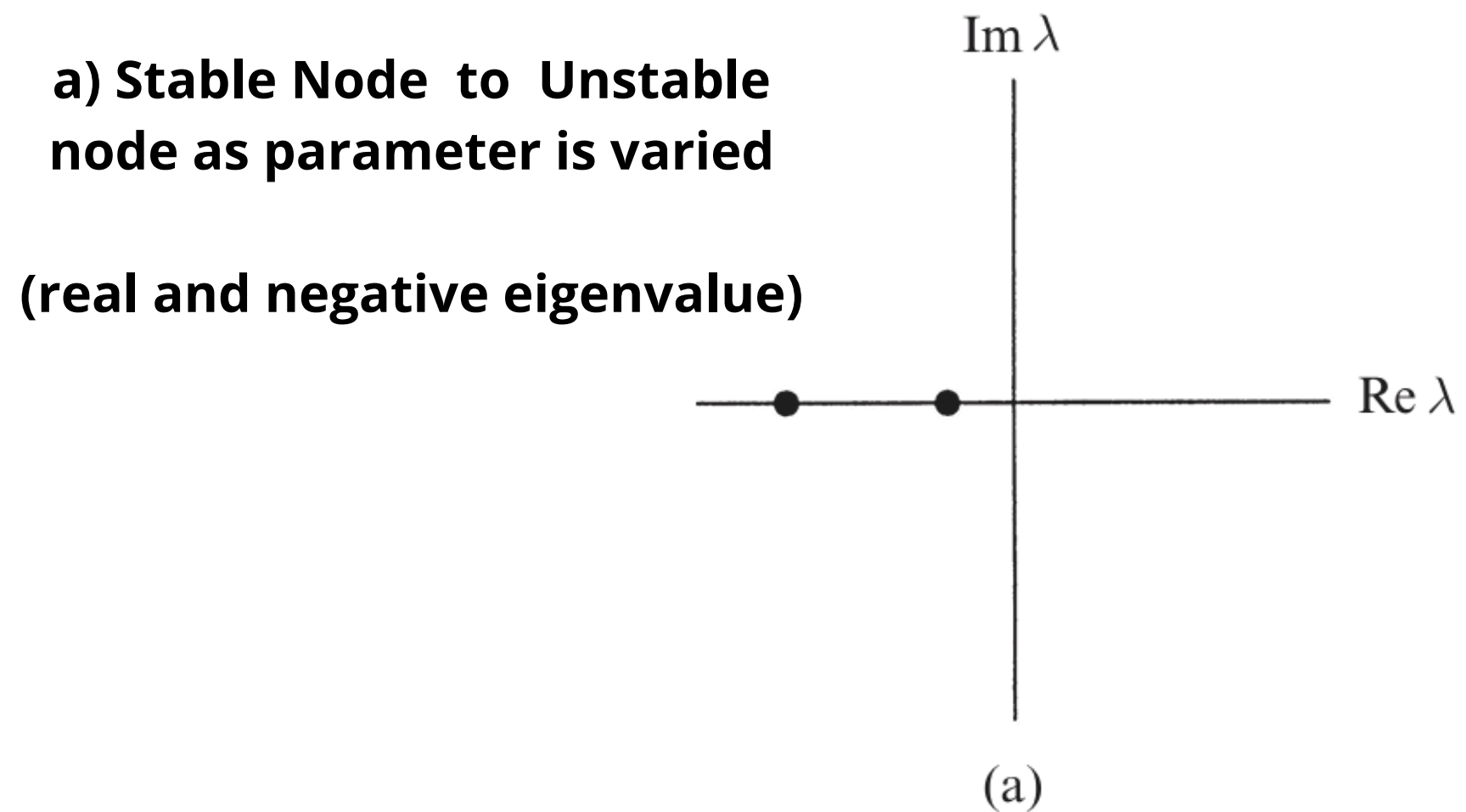


**Large
Deviations
when unable to
pass through**



The Math - Non Linear Dynamics

Hopf Bifurcation



$$\dot{x} = v \quad (1)$$

$$\begin{aligned} \dot{v} = k_c \left[n \cdot \sinh \left(\frac{v}{f_{ind}} \right) \right. \\ \left. - 2 \left(v + g \cdot \tanh \left(\frac{x}{\epsilon} \right) \right) \cosh \left(\frac{v}{f_{ind}} \right) \right] \\ - \frac{g}{\epsilon} v \cdot \operatorname{sech}^2 \left(\frac{x}{\epsilon} \right) \end{aligned} \quad (2)$$

the linearized system:

$$\begin{aligned} \dot{x} &= v \\ \dot{v} &= -\frac{2k_c g}{\epsilon} x + \left[k_c \left(\frac{n}{f_{ind}} - 2 \right) - \frac{g}{\epsilon} \right] v \end{aligned}$$

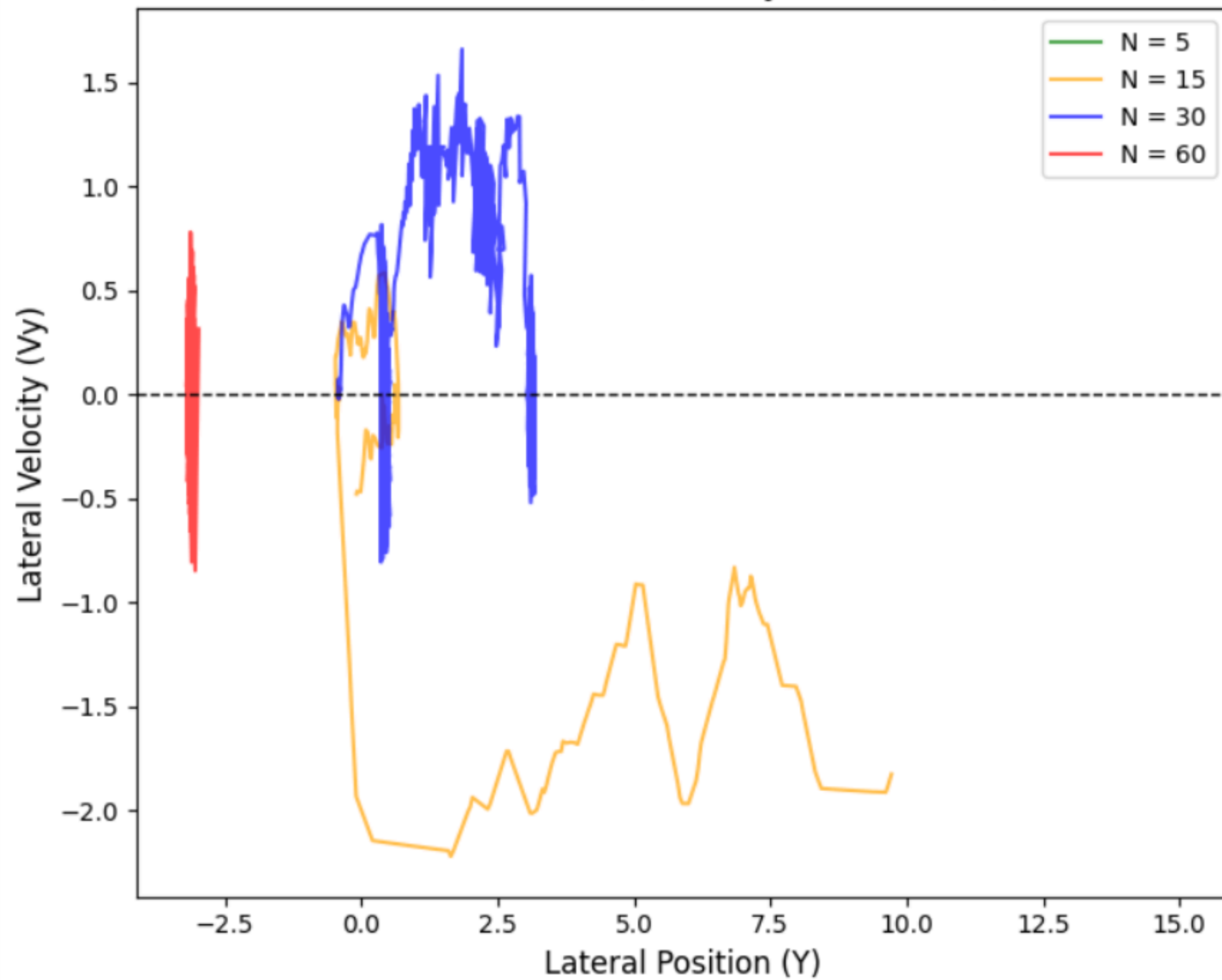
$$\begin{pmatrix} \dot{x} \\ \dot{v} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\omega^2 & \mu \end{pmatrix} \begin{pmatrix} x \\ v \end{pmatrix}$$

$$\begin{aligned} \mu &= k_c \left(\frac{n}{f_{ind}} - 2 \right) - \frac{g}{\epsilon} \\ \omega^2 &= \frac{2k_c g}{\epsilon} \end{aligned}$$

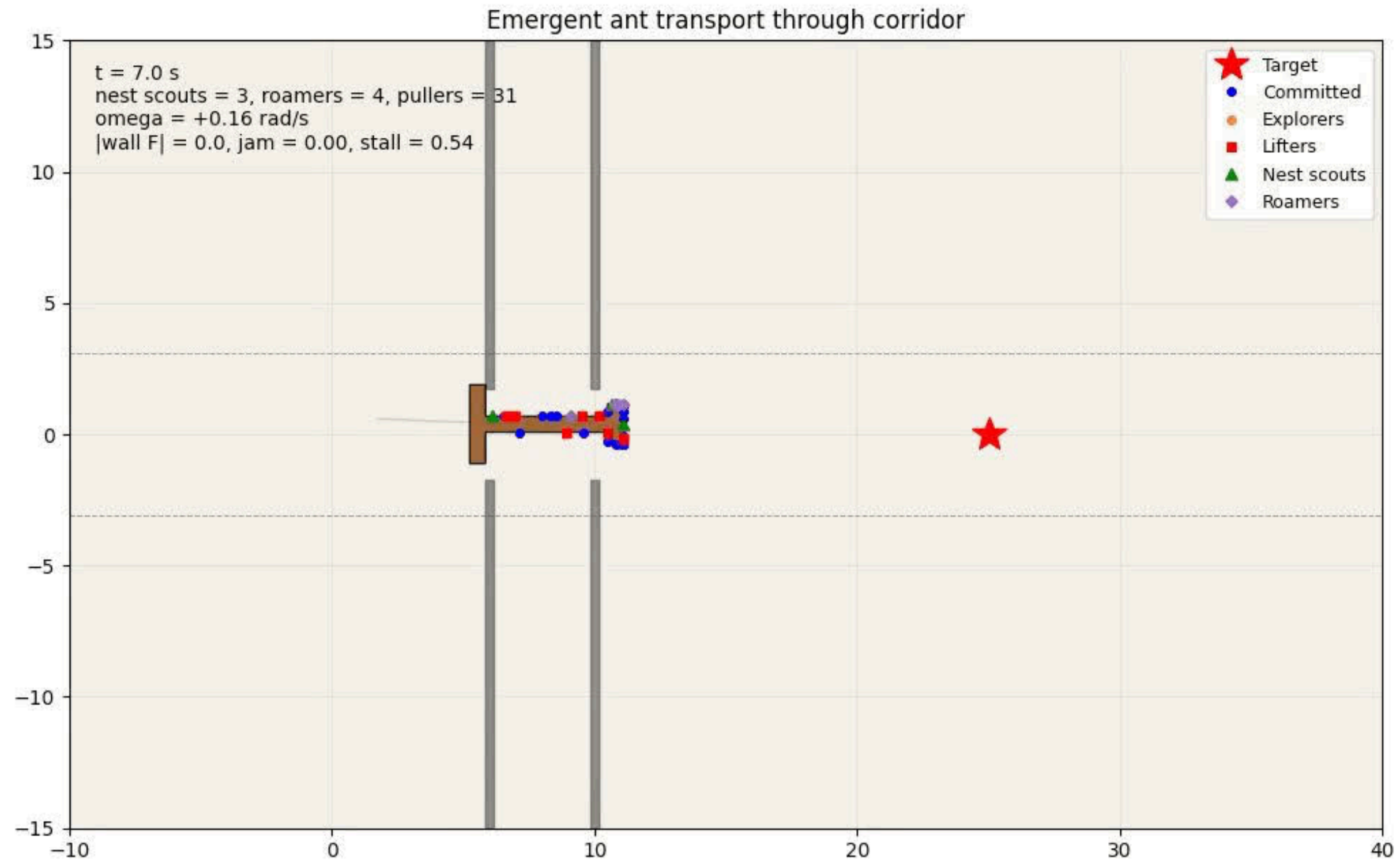
$$\lambda_{1,2} = \frac{1}{2} \left(\mu \pm \sqrt{\mu^2 - 4\omega^2} \right)$$

$$f_c^{(1)} = \frac{n}{2} \left(\frac{1}{1 + \frac{g}{2k_c \epsilon}} \right)$$

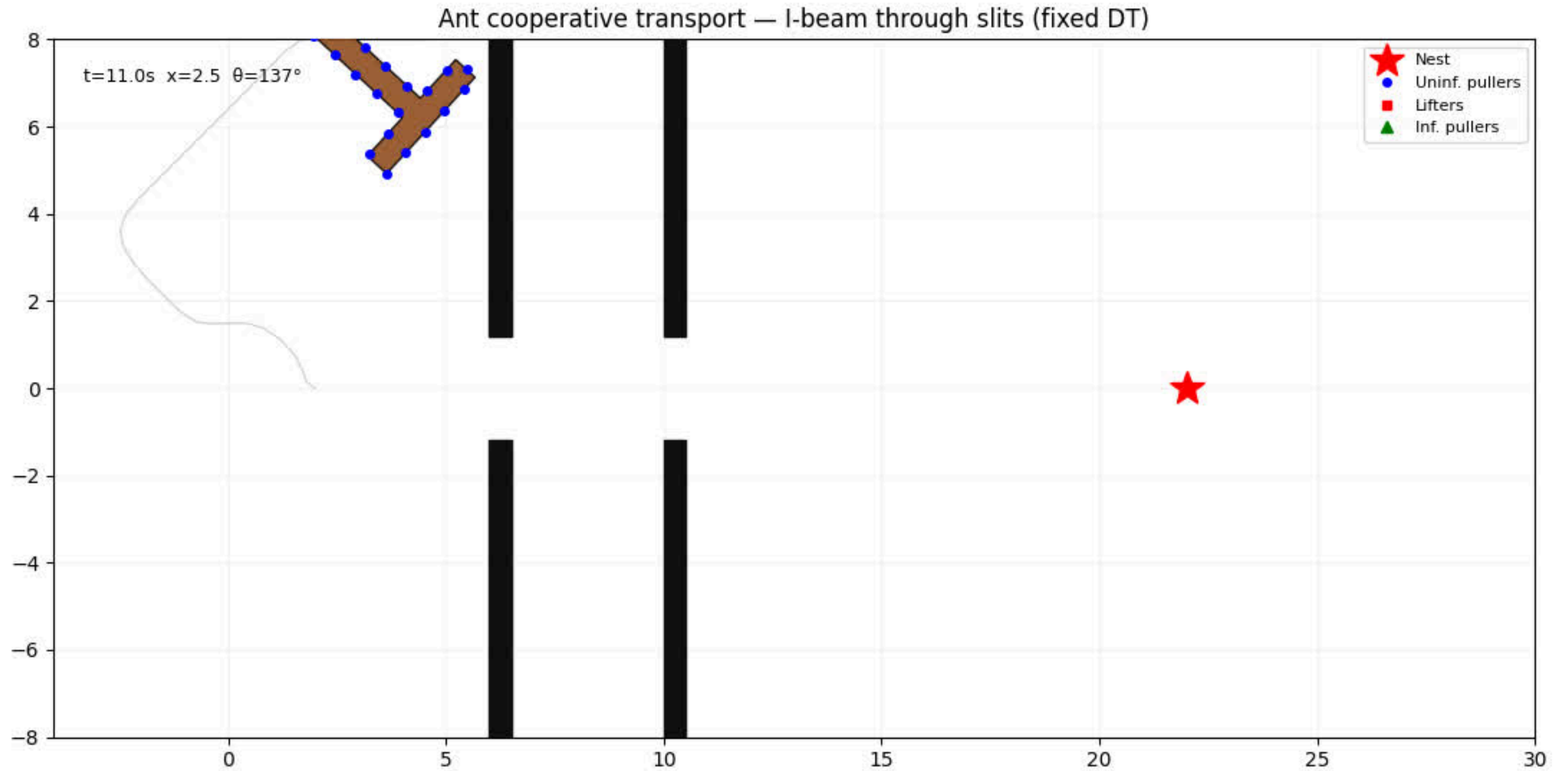
Phase Portraits (Velocity vs. Position)



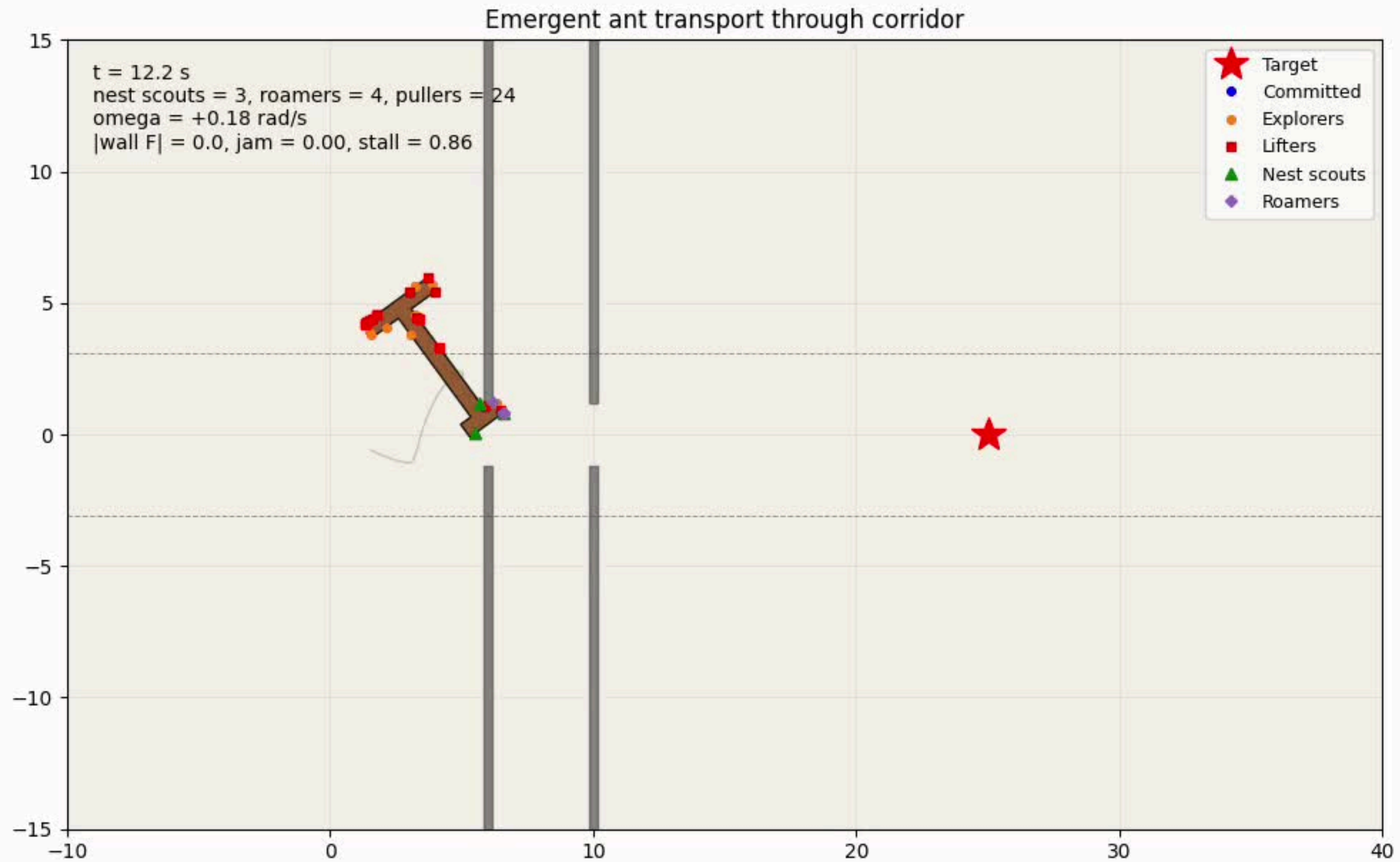
Base case - obstacle free board(wide slit) with a target



Tunelling issue through walls- Gillespie Time evolution



Best case simulation we could get



Problems and possible causes we tried out work on...

1. Beam sticking to the wall

Cause: Wall repulsion force lacked a velocity-damping or tangential component — the normal repulsion cancelled beam momentum but left it pinned at equilibrium distance with no mechanism to escape laterally.

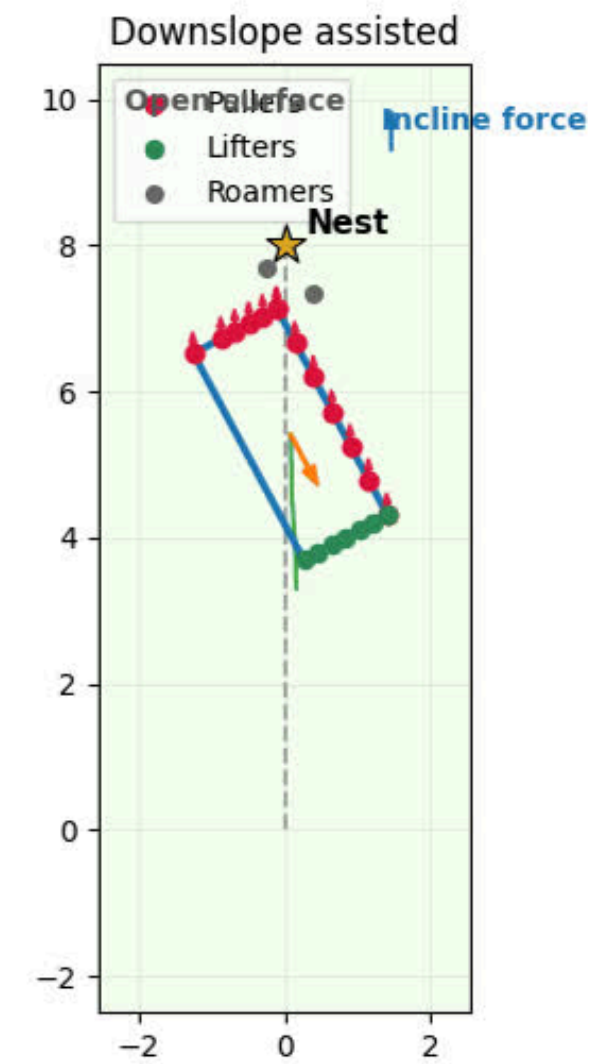
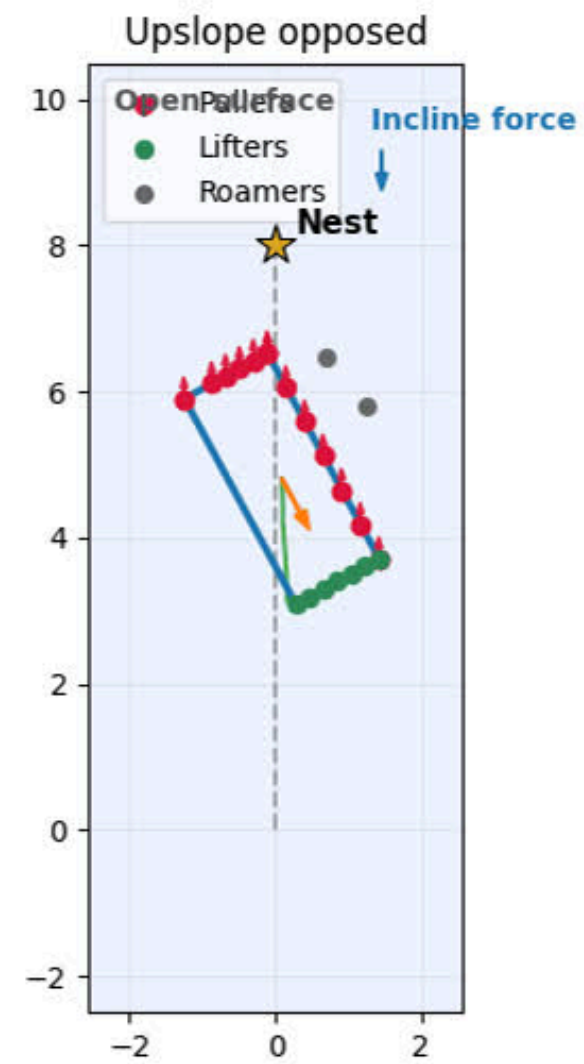
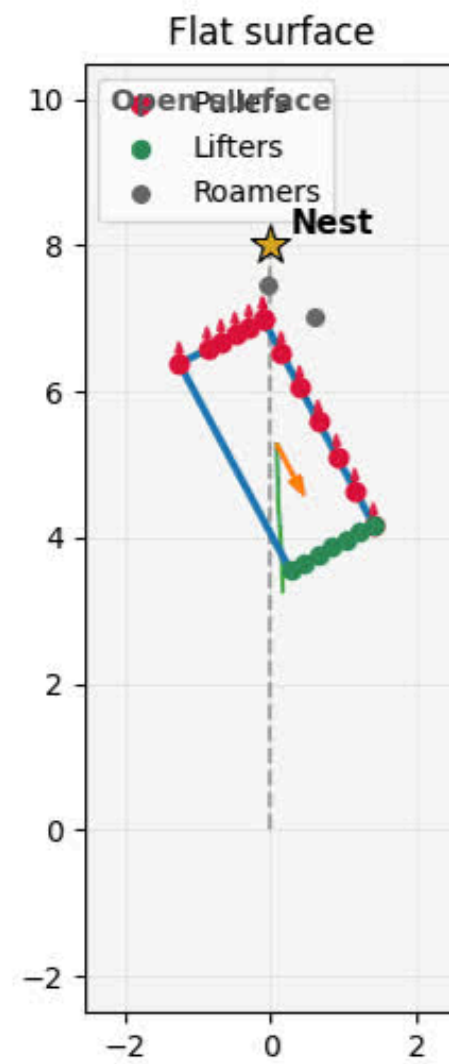
2. Beam tunnelling through vertical walls (Gillespie time evolution)

Cause: Gillespie algorithm selects discrete event times stochastically; if the time-step drawn is large, the beam position update jumps discontinuously across a wall boundary before collision detection fires — classical tunnelling from under-sampled spatial resolution at high drift velocities.

3. Random roaming before directed pulling (fewer informed pullers):

Cause: With low informed-puller count, the directional bias in the resultant force vector is statistically weak — thermal/stochastic kicks dominate over the signal, so the beam undergoes diffusion-like wandering until enough pullers accumulate to break symmetry.

Collective Transport on Slopes



$$\mathbf{F}_{\text{tot}} = \mathbf{F}_{\text{ants}} + \mathbf{F}_{\text{load}} + \mathbf{F}_{\text{slip}} + \mathbf{F}_{\text{contact}}$$

where:

- $\mathbf{F}_{\text{ants}}$ is the total pull generated by ants,
- $\mathbf{F}_{\text{load}}$ is the slope-induced load,
- $\mathbf{F}_{\text{slip}}$ is an additional downhill force when grip weakens,
- $\mathbf{F}_{\text{contact}}$ is a constraint/contact term when relevant.

$$\mathbf{F}_{\text{slip}} = -k_s \left(\frac{L - L_c}{L_c} \right) \left(1 - \frac{v}{v_c} \right) \hat{\mathbf{d}}$$

for $L > L_c$, and zero otherwise.

Here:

- k_s controls slip strength,
- L_c is the load threshold for slip onset,
- v_c is the speed scale below which slipping becomes severe.

Each active puller exerts a force

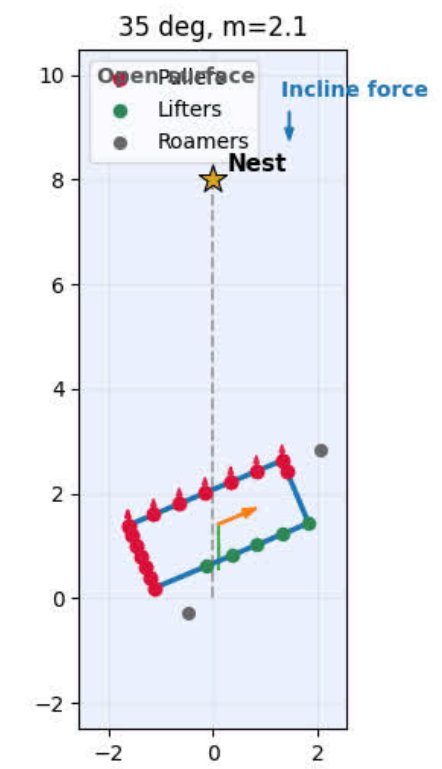
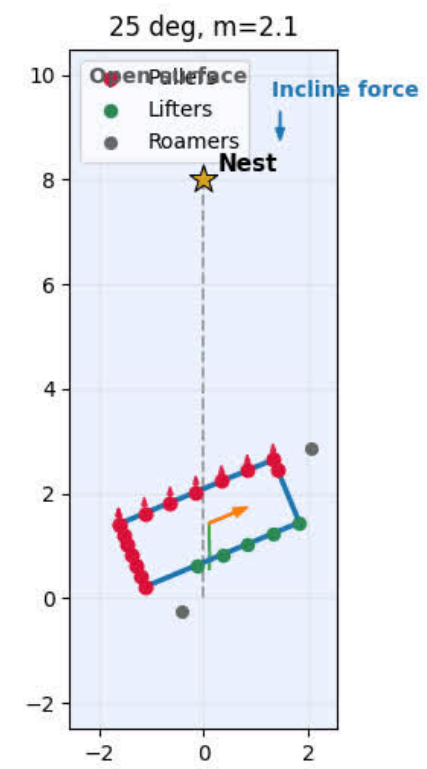
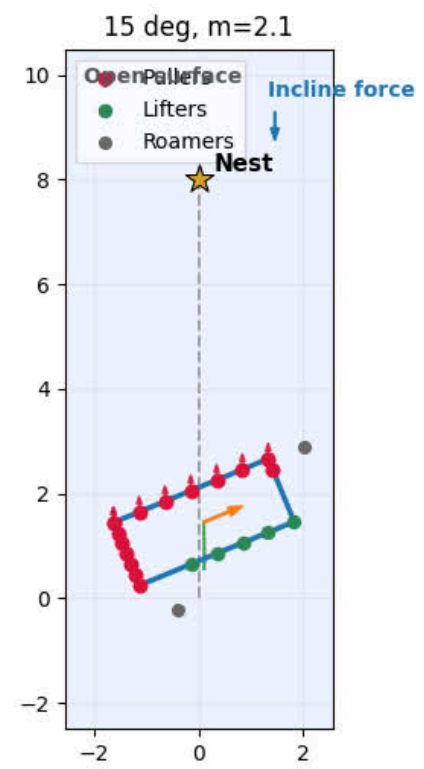
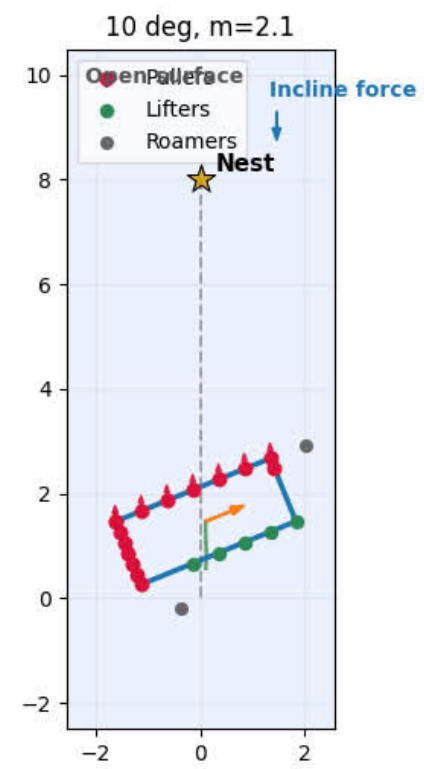
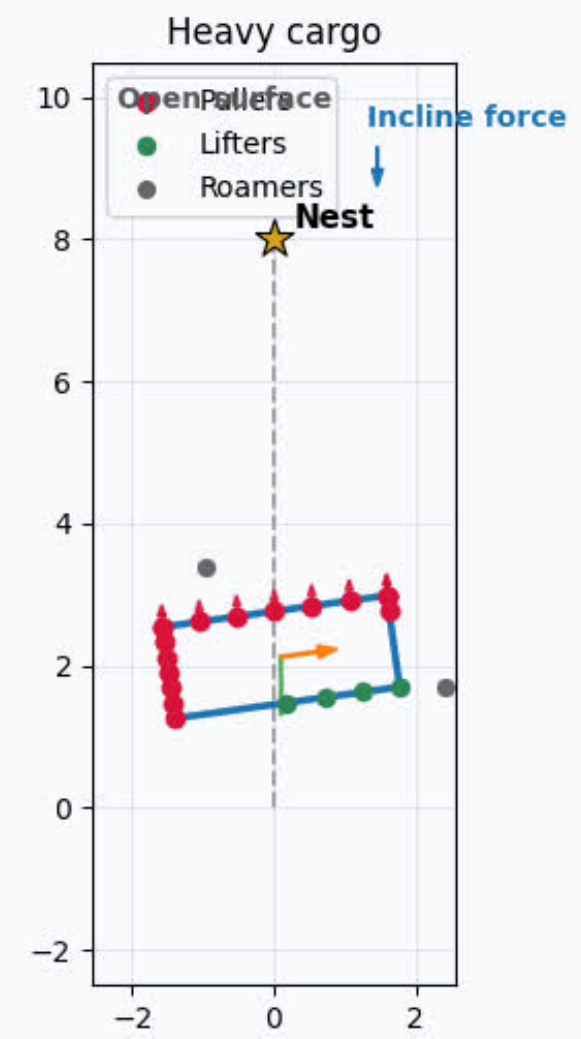
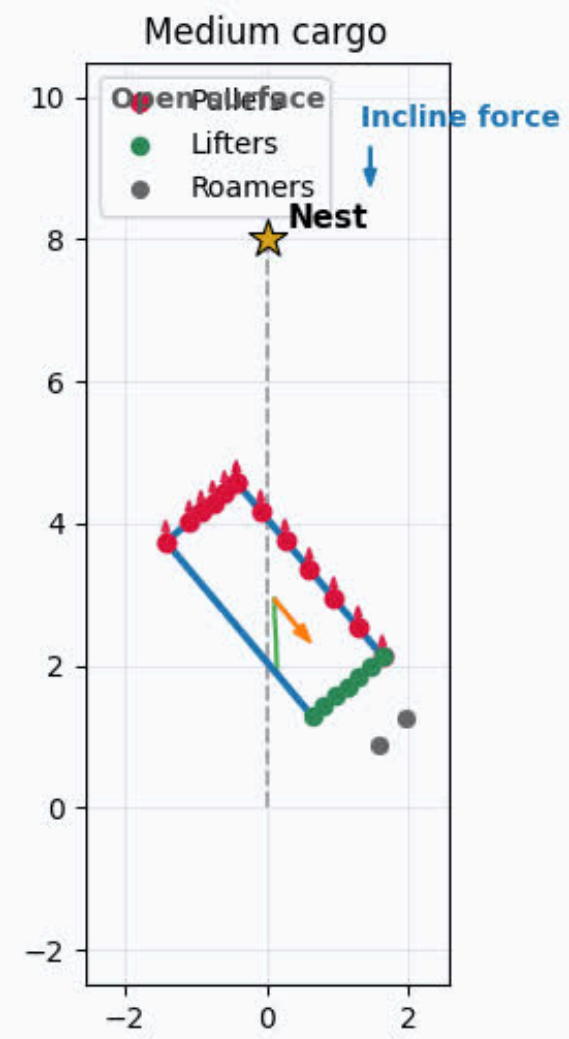
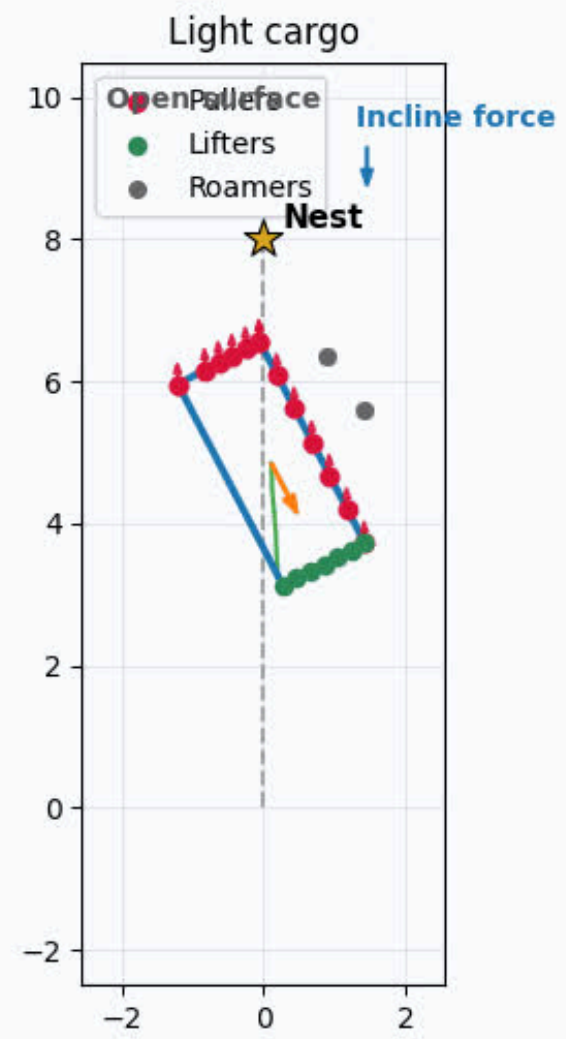
$$\mathbf{F}_i = F_p S_i \hat{\mathbf{d}}_i$$

where F_p is the baseline pulling strength, $S_i \in [0, 1]$ is a stamina factor, and $\hat{\mathbf{d}}_i$ is the pulling direction of ant i .

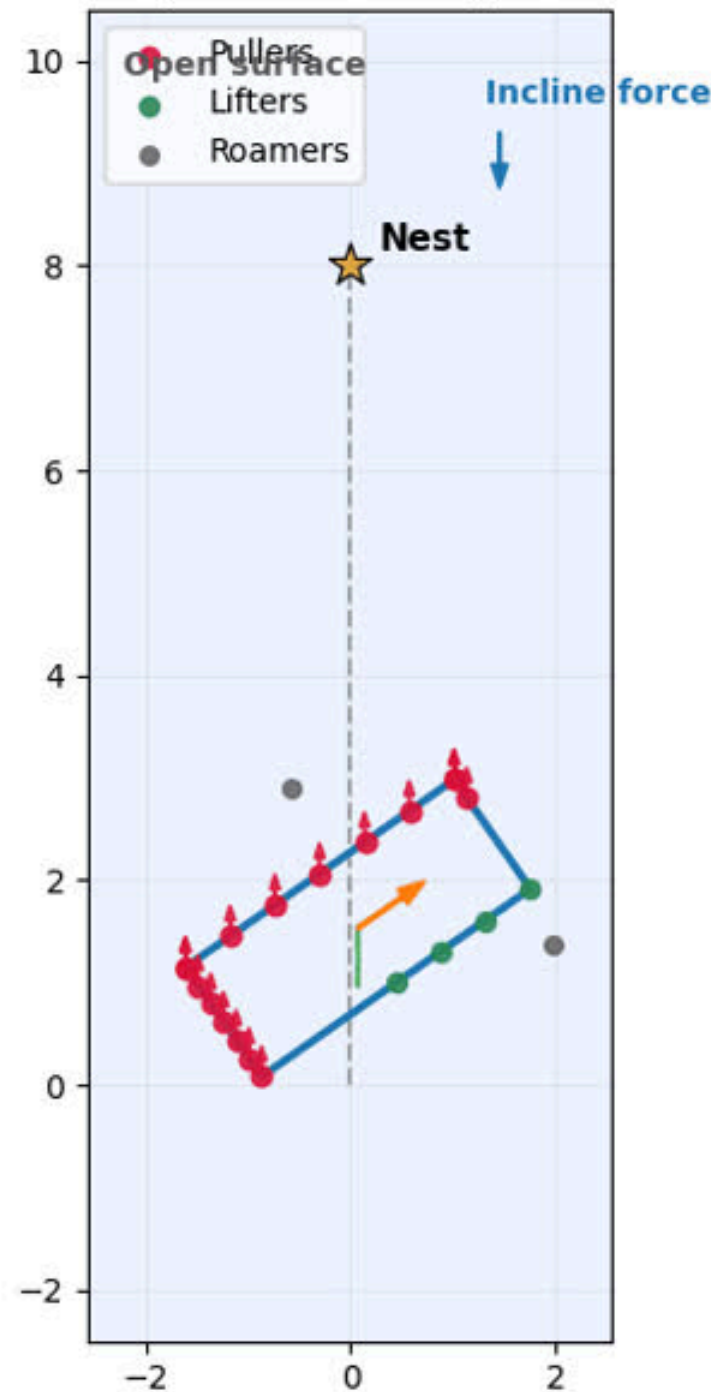
The total ant force is

$$\mathbf{F}_{\text{ants}} = \sum_{i \in \text{pullers}} \mathbf{F}_i$$

Thus, collective pulling depends on both the number of pullers and the directional agreement among them.



Inclined case: upward motion opposed, frame 094



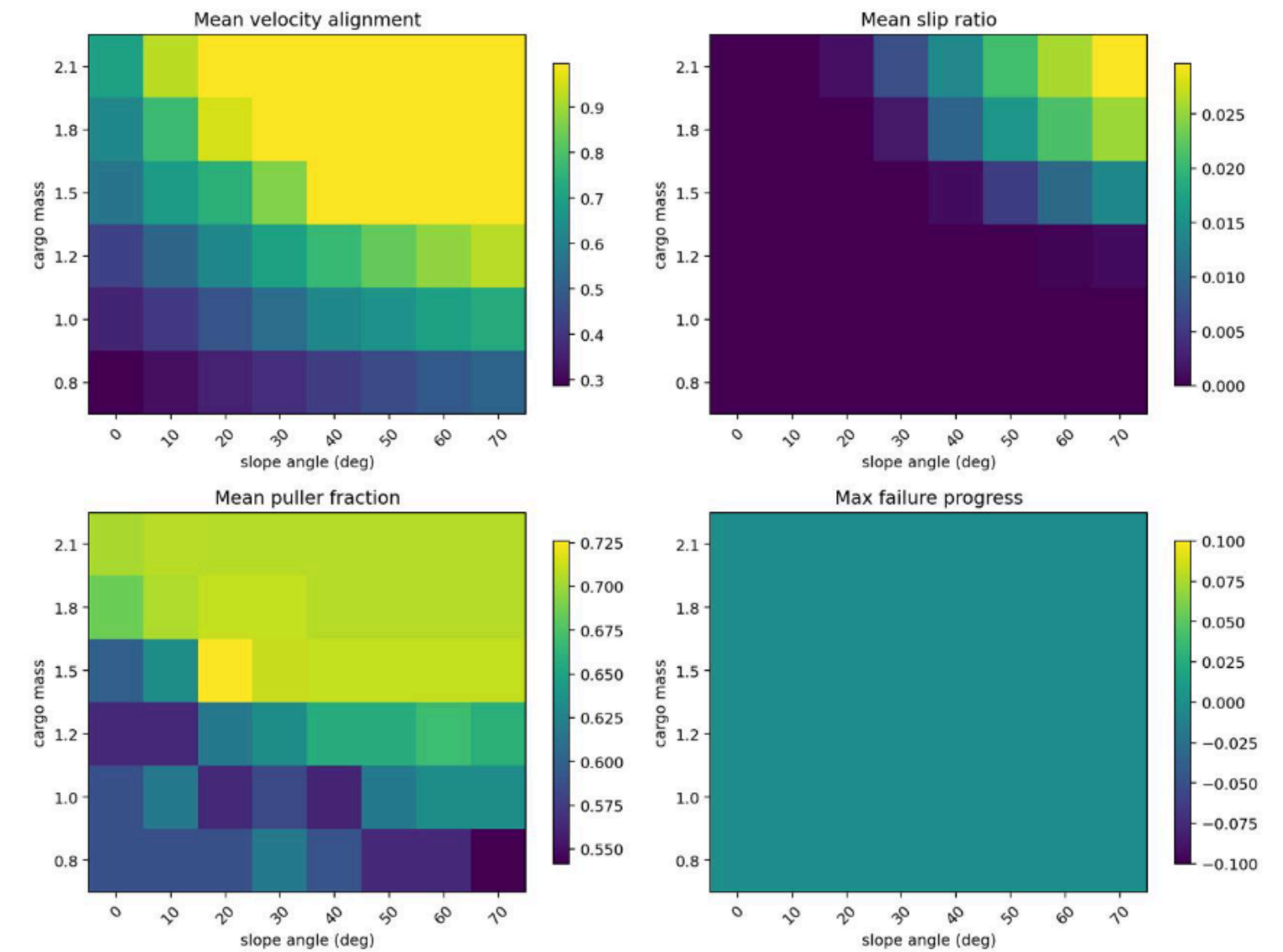
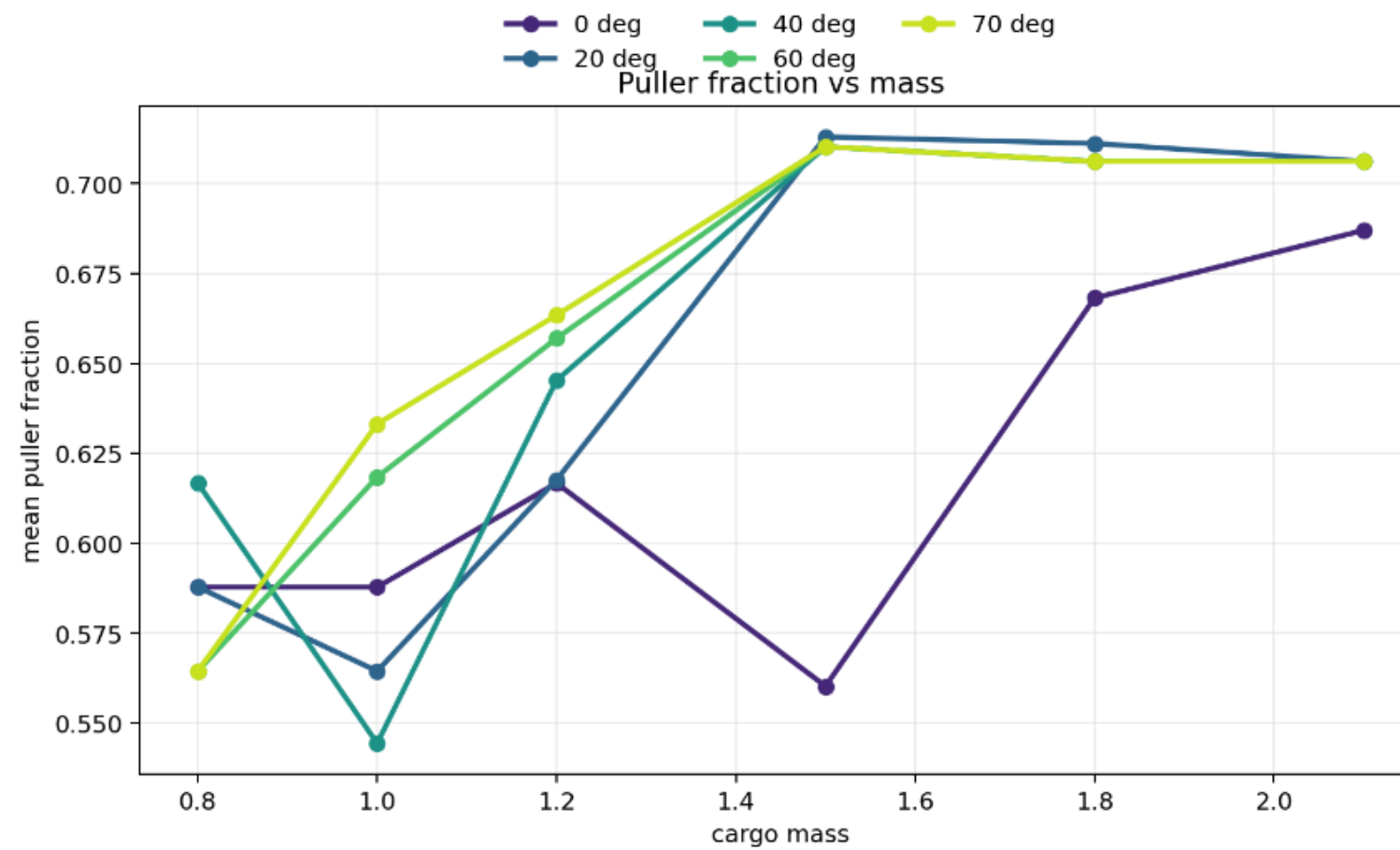
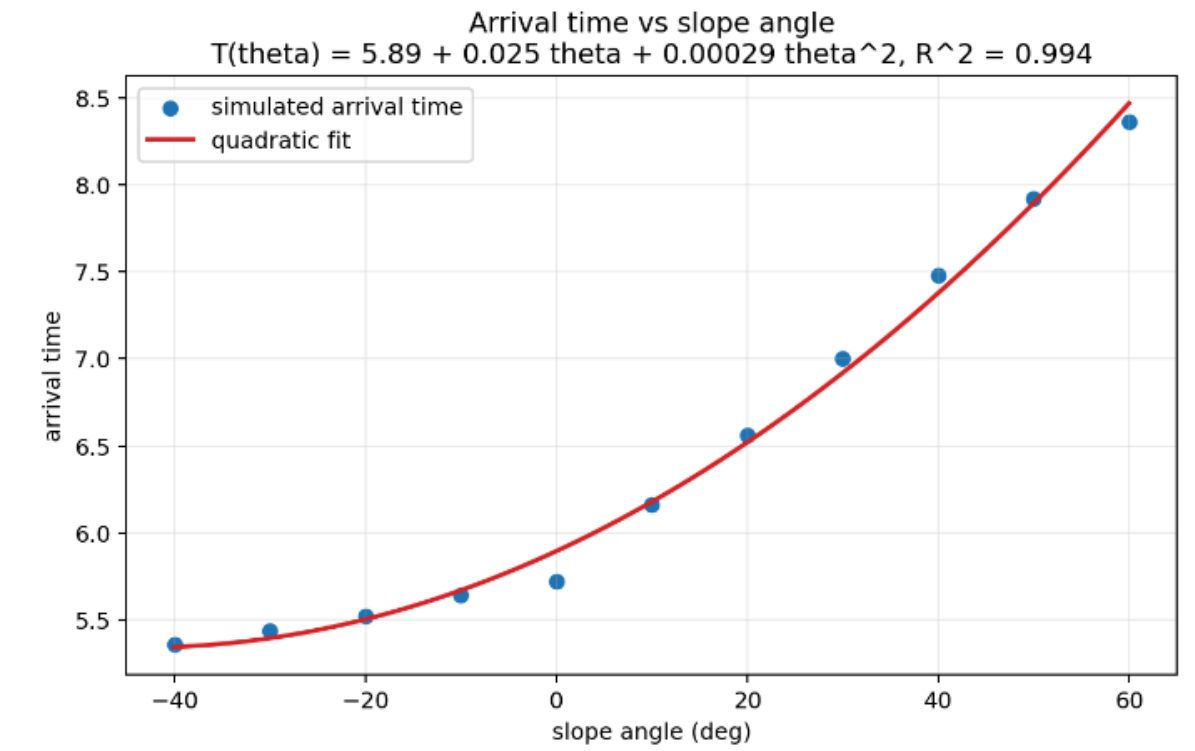
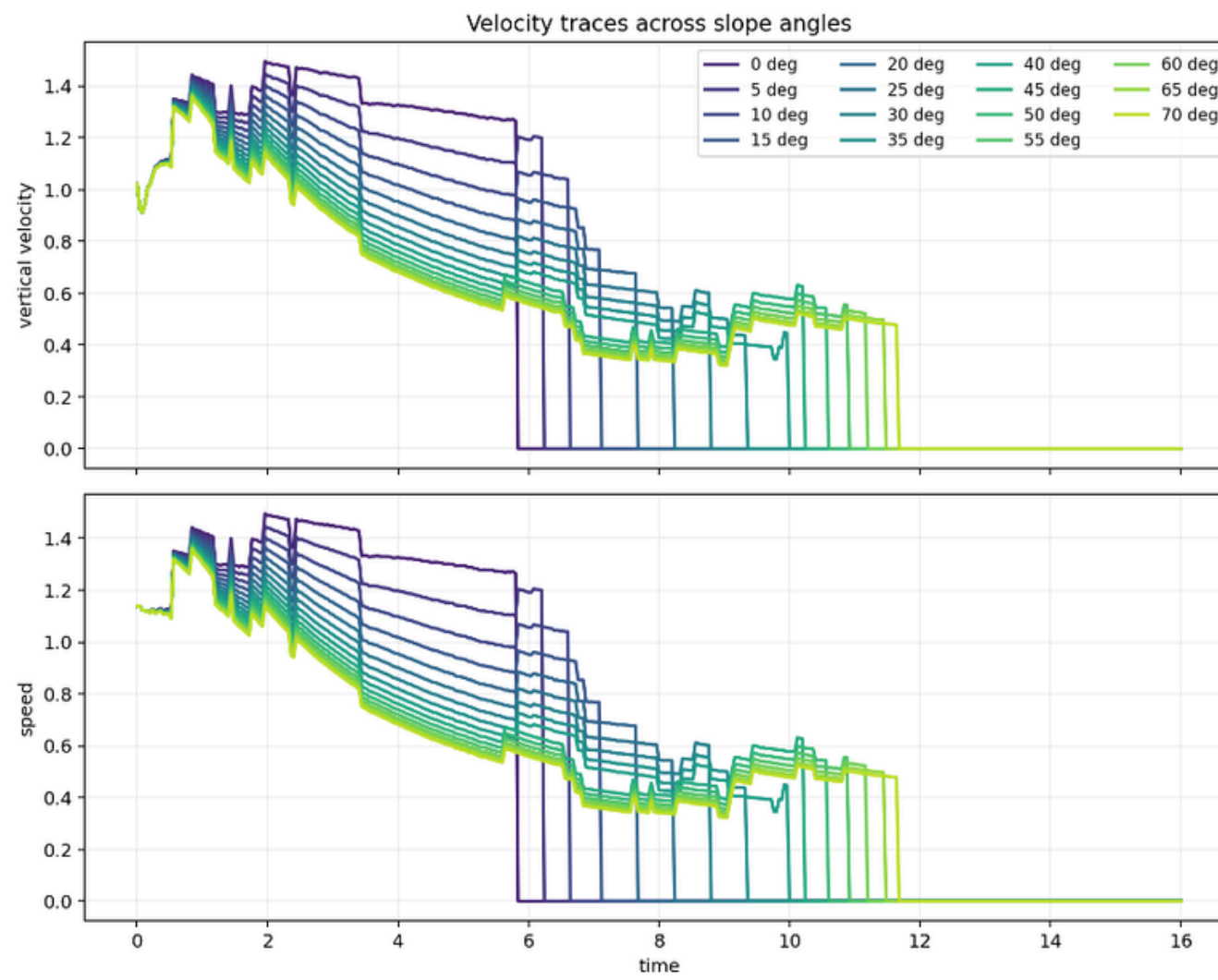
Each puller loses stamina over time according to

$$S_i(t + \Delta t) = S_i(t) - k_{\text{fatigue}}\Delta t$$

while non-pulling ants recover according to

$$S_i(t + \Delta t) = S_i(t) + k_{\text{recovery}}\Delta t$$

When stamina falls sufficiently low, pullers are reassigned to less demanding roles such as lifter or roamer. Conversely, sufficiently recovered ants can re-enter active pulling. This enables dynamic redistribution of labor during transport.



Thank You